

Derivatives

Derivatives and Rates of Change

Derivatives and Rates of Change

The problem of finding the tangent line to a curve and the problem of finding the velocity of an object both involve finding the same type of limit.

This special type of limit is called a *derivative* and we will see that it can be interpreted as a rate of change in any of the sciences or engineering.

Tangents

Tangents

If a curve C has equation $y = f(x)$ and we want to find the tangent line to C at the point $P(a, f(a))$, then we consider a nearby point $Q(x, f(x))$, where $x \neq a$, and compute the slope of the secant line PQ :

$$m_{PQ} = \frac{f(x) - f(a)}{x - a}$$

Then we let Q approach P along the curve C by letting x approach a .

Tangents

If m_{PQ} approaches a number m , then we define the *tangent* t to be the line through P with slope m . (This amounts to saying that the tangent line is the limiting position of the secant line PQ as Q approaches P . See Figure 1.)

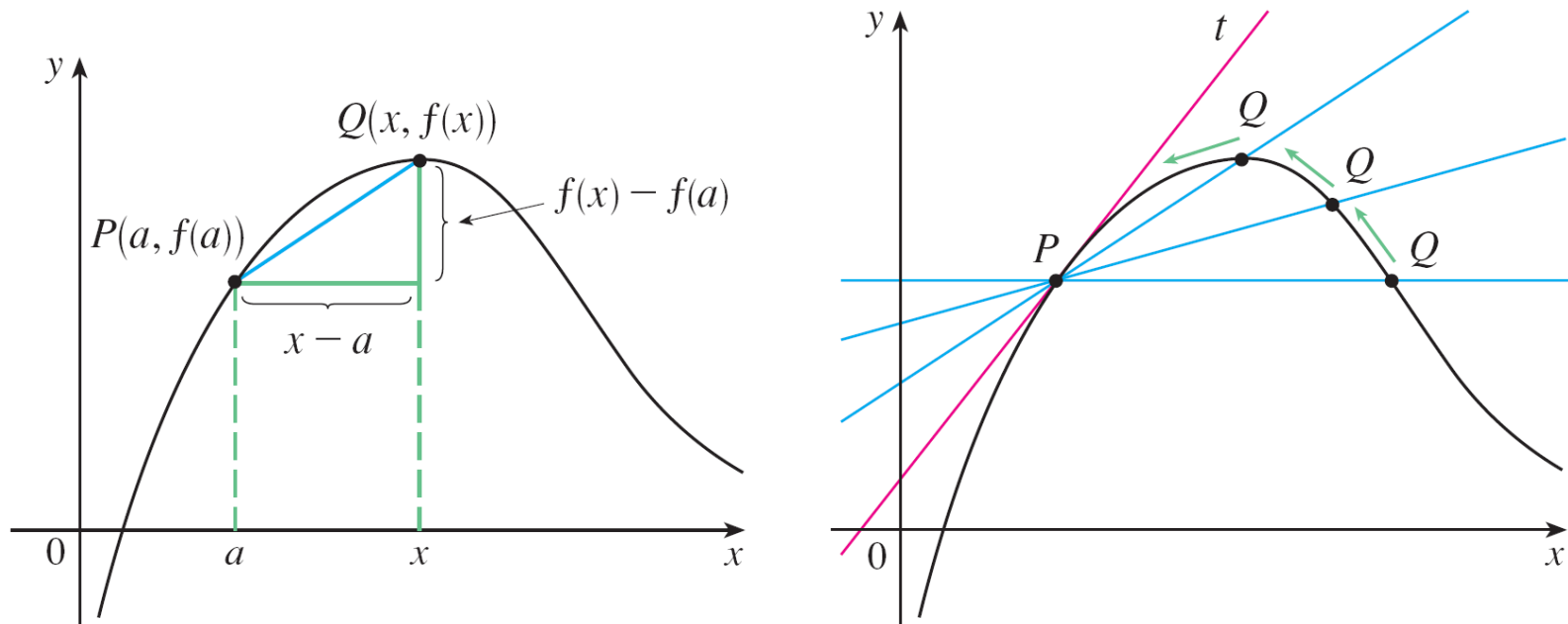


Figure 1

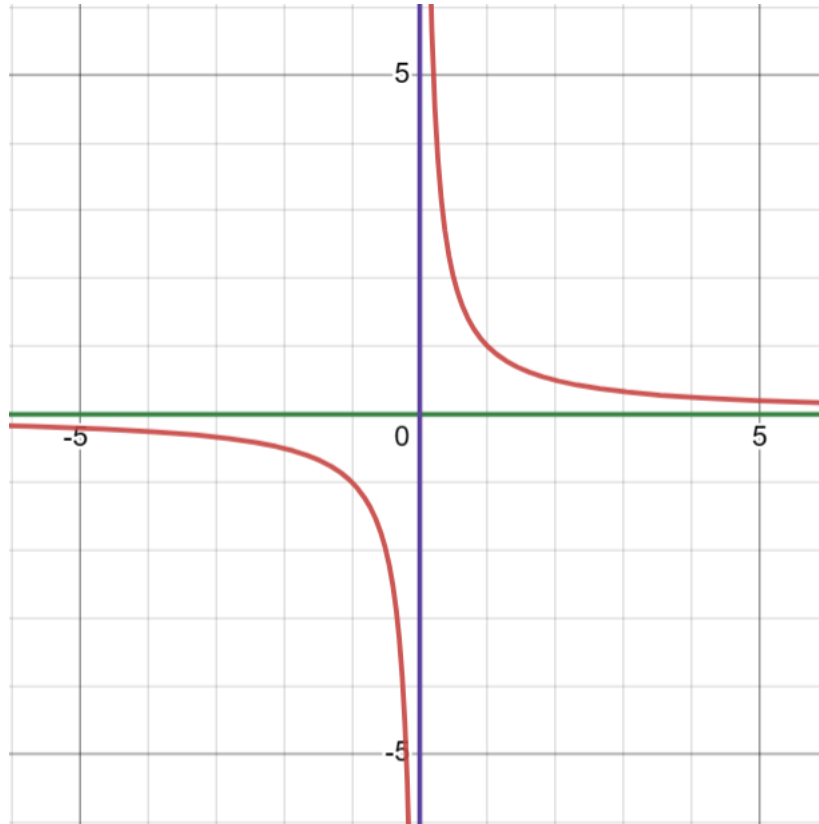
Tangents

1 Definition The **tangent line** to the curve $y = f(x)$ at the point $P(a, f(a))$ is the line through P with slope

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

provided that this limit exists.

Asymptotes



$$F(x) = 1/x$$

An asymptote is a type of restriction on the domain or range of a given function. Graphically, an asymptote looks like a value line that a function gets infinitely close to but never actually reaches.

Example 1

Find an equation of the tangent line to the parabola $y = x^2$ at the point $P(1, 1)$.

Solution:

Here we have $a = 1$ and $f(x) = x^2$, so the slope is

$$\begin{aligned} m &= \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} \\ &= \lim_{x \rightarrow 1} \frac{(x - 1)(x + 1)}{x - 1} \\ &= \lim_{x \rightarrow 1} (x + 1) \end{aligned}$$

Example 1 – *Solution*

cont'd

$$= 1 + 1$$

$$= 2$$

Using the point-slope form of the equation of a line, we find that an equation of the tangent line at $(1, 1)$ is

$$y - 1 = 2(x - 1) \quad \text{or} \quad y = 2x - 1$$

Tangents

If $h = x - a$, then $x = a + h$ and so the slope of the secant line PQ is

$$m_{PQ} = \frac{f(a + h) - f(a)}{h}$$

(See Figure 3 where the case $h > 0$ is illustrated and Q is to the right of P . If it happened that $h < 0$, however, Q would be to the left of P .)

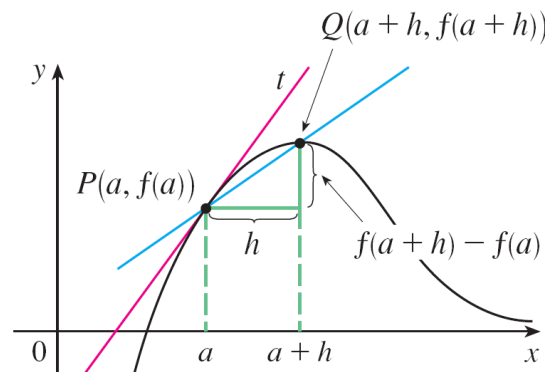


Figure 3

Tangents

Notice that as x approaches a , h approaches 0 (because $h = x - a$) and so the expression for the slope of the tangent line in Definition 1 becomes

2

$$m = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

Velocities

Velocities

In general, suppose an object moves along a straight line according to an equation of motion $s = f(t)$, where s is the displacement (directed distance) of the object from the origin at time t .

The function f that describes the motion is called the **position function** of the object. In the time interval from $t = a$ to $t = a + h$ the change in position is $f(a + h) - f(a)$. (See Figure 5.)

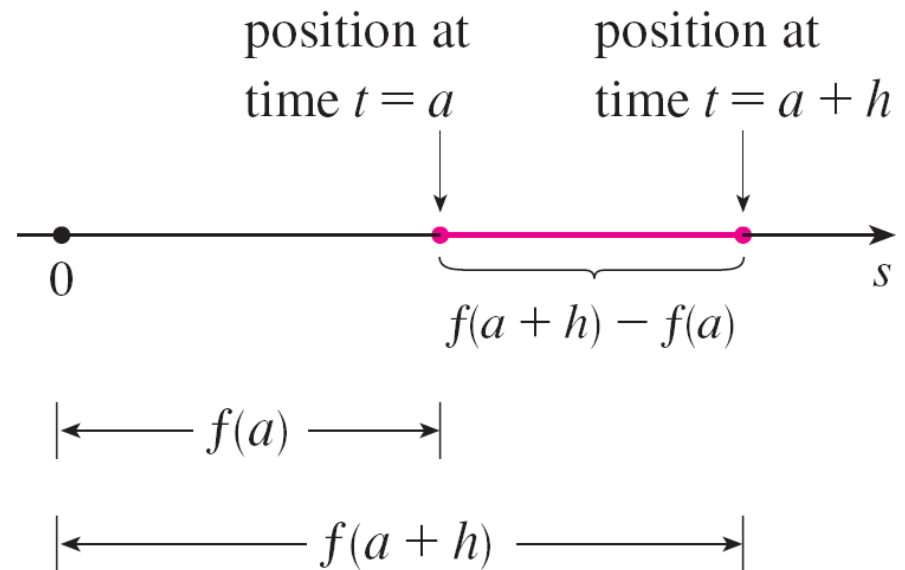


Figure 5

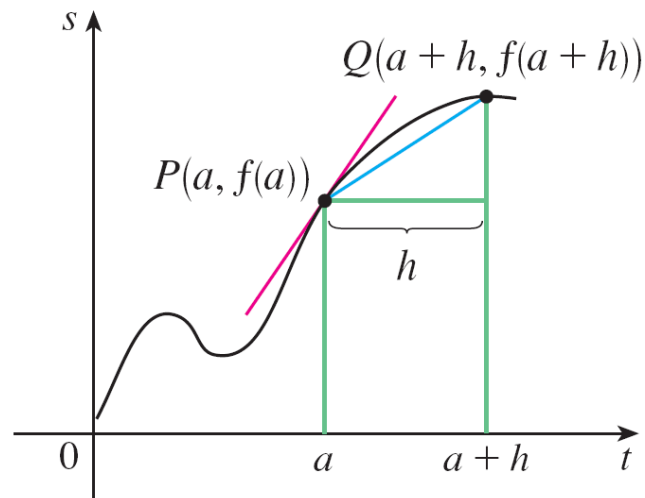
Velocities

The average velocity over this time interval is

$$\text{average velocity} = \frac{\text{displacement}}{\text{time}} = \frac{f(a + h) - f(a)}{h}$$

Velocities

which is the same as the slope of the secant line PQ in Figure 6.



$$\begin{aligned} m_{PQ} &= \frac{f(a+h) - f(a)}{h} \\ &= \text{average velocity} \end{aligned}$$

Figure 6

Velocities

Now suppose we compute the average velocities over shorter and shorter time intervals $[a, a + h]$.

In other words, we let h approach 0. As in the example of the falling ball, we define the **velocity** (or **instantaneous velocity**) $v(a)$ at time $t = a$ to be the limit of these average velocities:

3

$$v(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

This means that the velocity at time $t = a$ is equal to the slope of the tangent line at P .

Example 3

Suppose that a ball is dropped from the upper observation deck of the CN Tower, 450 m above the ground.

- (a) What is the velocity of the ball after 5 seconds?
- (b) How fast is the ball traveling when it hits the ground?

Solution:

We will need to find the velocity both when $t = 5$ and when the ball hits the ground, so it's efficient to start by finding the velocity at a general time $t = a$.

Example 3 – *Solution*

cont'd

Using the equation of motion $s = f(t) = 4.9t^2$, we have

$$\begin{aligned}v(a) &= \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h} \\&= \lim_{h \rightarrow 0} \frac{4.9(a + h)^2 - 4.9a^2}{h} \\&= \lim_{h \rightarrow 0} \frac{4.9(a^2 + 2ah + h^2 - a^2)}{h} \\&= \lim_{h \rightarrow 0} \frac{4.9(2ah + h^2)}{h} \\&= \lim_{h \rightarrow 0} 4.9(2a + h) \\&= 9.8a\end{aligned}$$

Example 3 – *Solution*

cont'd

(a) The velocity after 5 s is $v(5) = (9.8)(5) = 49$ m/s.

(b) Since the observation deck is 450 m above the ground, the ball will hit the ground at the time t_1 when $s(t_1) = 450$, that is,

$$4.9t_1^2 = 450$$

This gives

$$t_1^2 = \frac{450}{4.9} \quad \text{and} \quad t_1 = \sqrt{\frac{450}{4.9}} \approx 9.6 \text{ s}$$

The velocity of the ball as it hits the ground is therefore

$$v(t_1) = 9.8t_1 = 9.8\sqrt{\frac{450}{4.9}} \approx 94 \text{ m/s}$$

Derivatives

Derivatives

We have seen that the same type of limit arises in finding the slope of a tangent line (Equation 2) or the velocity of an object (Equation 3).

In fact, limits of the form

$$\lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

arise whenever we calculate a rate of change in any of the sciences or engineering, such as a rate of reaction in chemistry or a marginal cost in economics.

Since this type of limit occurs so widely, it is given a special name and notation.

Derivatives

4 Definition The **derivative of a function f at a number a** , denoted by $f'(a)$, is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

if this limit exists.

If we write $x = a + h$, then we have $h = x - a$ and h approaches 0 if and only if x approaches a . Therefore an equivalent way of stating the definition of the derivative, as we saw in finding tangent lines, is

5

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

Example 4

Find the derivative of the function $f(x) = x^2 - 8x + 9$ at the number a .

Solution:

From Definition 4 we have

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h} \\ &= \lim_{h \rightarrow 0} \frac{[(a + h)^2 - 8(a + h) + 9] - [a^2 - 8a + 9]}{h} \\ &= \lim_{h \rightarrow 0} \frac{a^2 + 2ah + h^2 - 8a - 8h + 9 - a^2 + 8a - 9}{h} \end{aligned}$$

Example 4 – *Solution*

cont'd

$$= \lim_{h \rightarrow 0} \frac{2ah + h^2 - 8h}{h}$$

$$= \lim_{h \rightarrow 0} (2a + h - 8)$$

$$= 2a - 8$$

Derivatives

We defined the tangent line to the curve $y = f(x)$ at the point $P(a, f(a))$ to be the line that passes through P and has slope m given by Equation 1 or 2.

Since, by Definition 4, this is the same as the derivative $f'(a)$, we can now say the following.

The tangent line to $y = f(x)$ at $(a, f(a))$ is the line through $(a, f(a))$ whose slope is equal to $f'(a)$, the derivative of f at a .

Derivatives

If we use the point-slope form of the equation of a line, we can write an equation of the tangent line to the curve $y = f(x)$ at the point $(a, f(a))$:

$$y - f(a) = f'(a)(x - a)$$

Rates of Change

Rates of Change

Suppose y is a quantity that depends on another quantity x . Thus y is a function of x and we write $y = f(x)$. If x changes from x_1 to x_2 , then the change in x (also called the **increment** of x) is

$$\Delta x = x_2 - x_1$$

and the corresponding change in y is

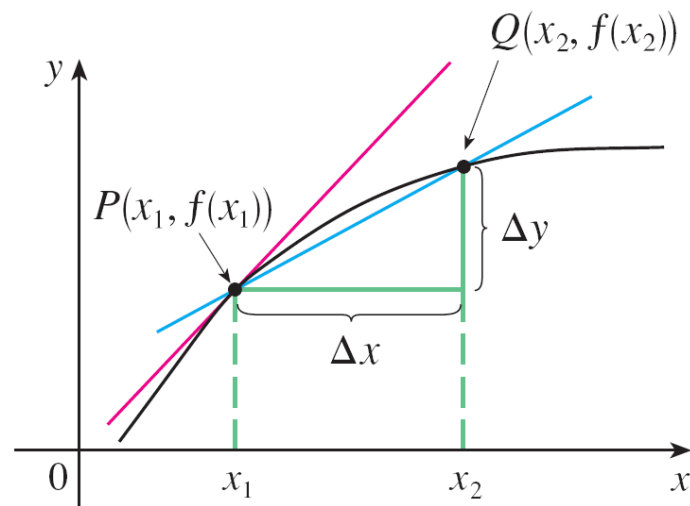
$$\Delta y = f(x_2) - f(x_1)$$

Rates of Change

The difference quotient

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

is called the **average rate of change of y with respect to x** over the interval $[x_1, x_2]$ and can be interpreted as the slope of the secant line PQ in Figure 8.



average rate of change = m_{PQ}

instantaneous rate of change =
slope of tangent at P

Figure 8

Rates of Change

By analogy with velocity, we consider the average rate of change over smaller and smaller intervals by letting x_2 approach x_1 and therefore letting Δx approach 0.

The limit of these average rates of change is called the **(instantaneous) rate of change of y with respect to x** at $x = x_1$, which is interpreted as the slope of the tangent to the curve $y = f(x)$ at $P(x_1, f(x_1))$:

$$\boxed{6} \quad \text{instantaneous rate of change} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{x_2 \rightarrow x_1} \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

We recognize this limit as being the derivative $f'(x_1)$.

Rates of Change

We know that one interpretation of the derivative $f'(a)$ is as the slope of the tangent line to the curve $y = f(x)$ when $x = a$. We now have a second interpretation:

The derivative $f'(a)$ is the instantaneous rate of change of $y = f(x)$ with respect to x when $x = a$.

The connection with the first interpretation is that if we sketch the curve $y = f(x)$, then the instantaneous rate of change is the slope of the tangent to this curve at the point where $x = a$.

Rates of Change

This means that when the derivative is large (and therefore the curve is steep, as at the point P in Figure 9), the y -values change rapidly.

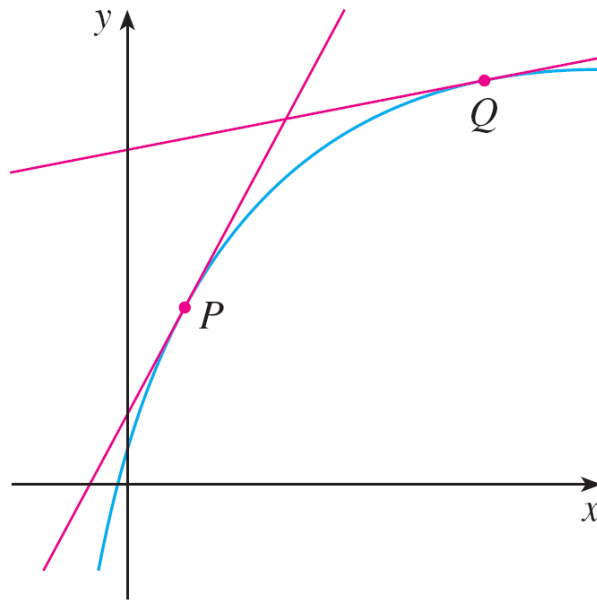


Figure 9

The y -values are changing rapidly at P and slowly at Q .

Rates of Change

When the derivative is small, the curve is relatively flat (as at point Q) and the y -values change slowly.

In particular, if $s = f(t)$ is the position function of a particle that moves along a straight line, then $f'(a)$ is the rate of change of the displacement s with respect to the time t .

In other words, $f'(a)$ is the velocity of the particle at time $t = a$.

The **speed** of the particle is the absolute value of the velocity, that is, $|f'(a)|$.